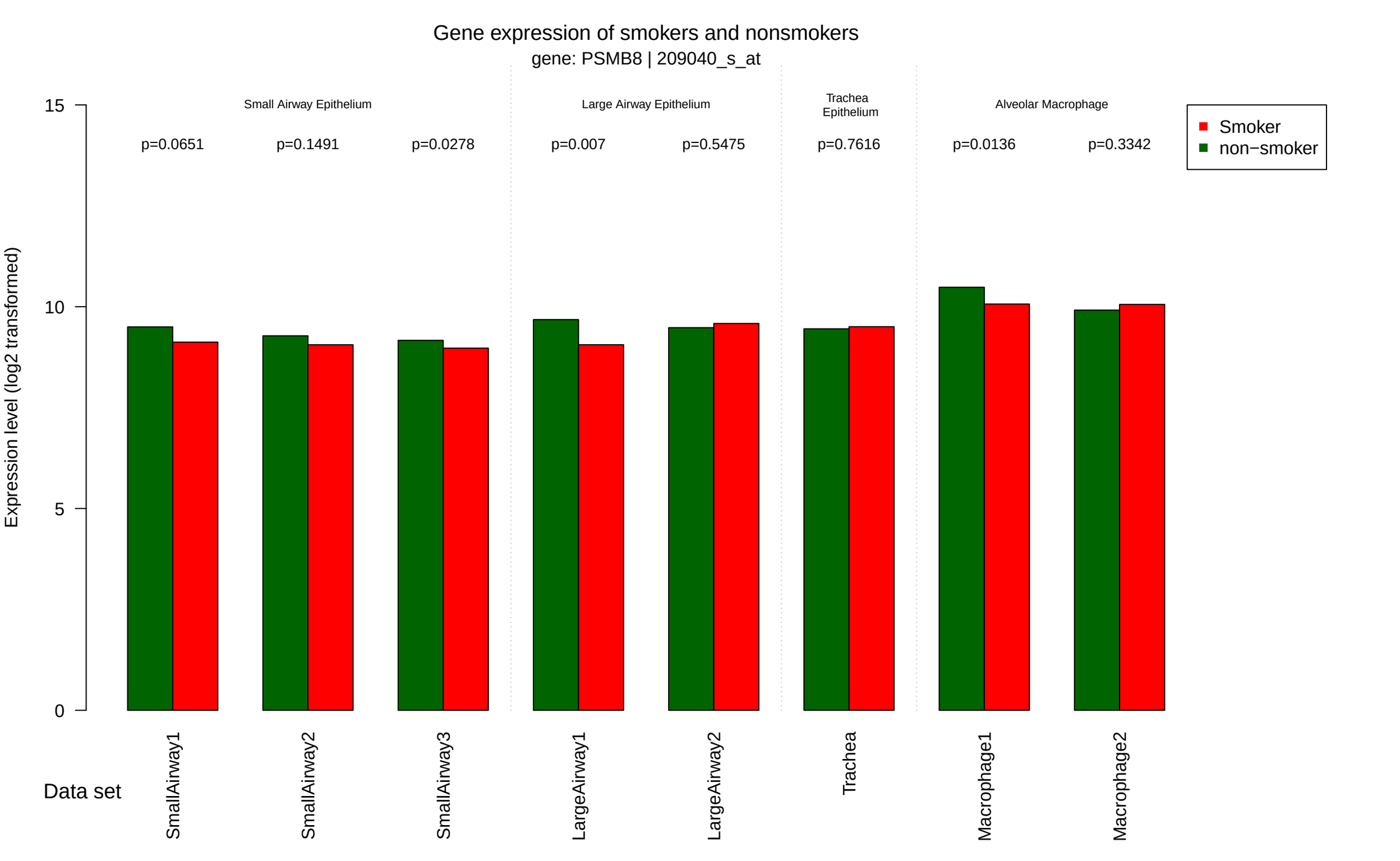
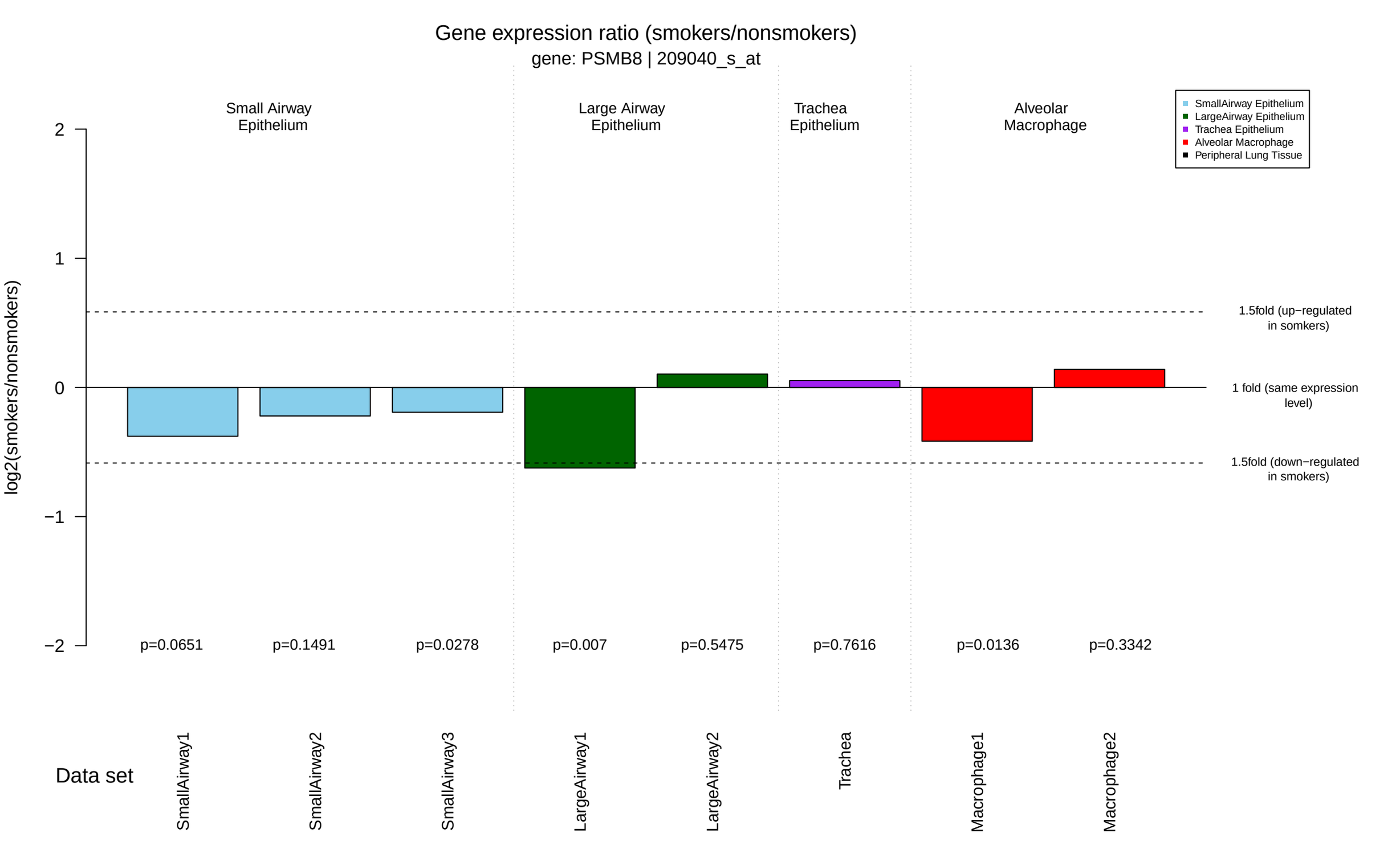
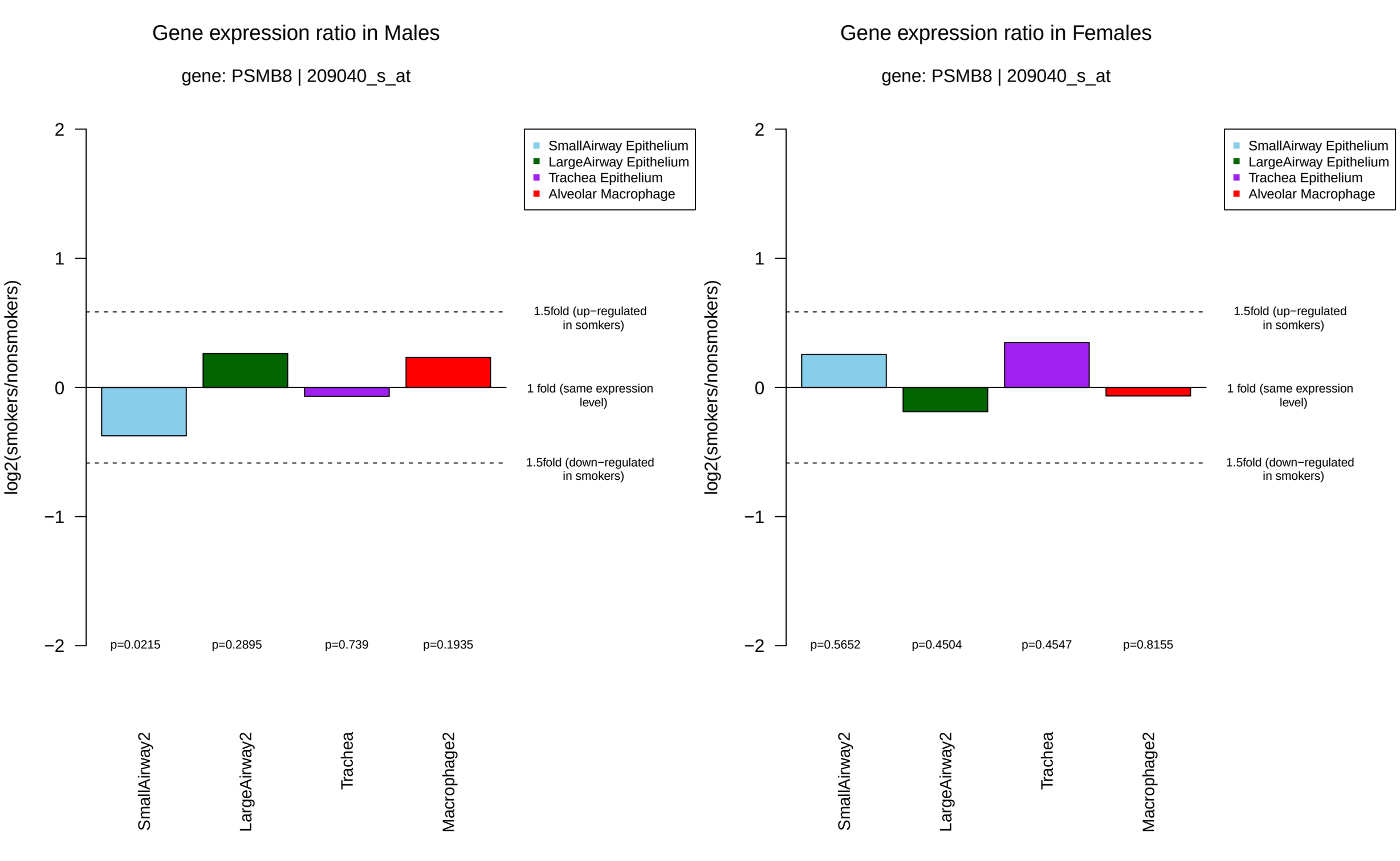
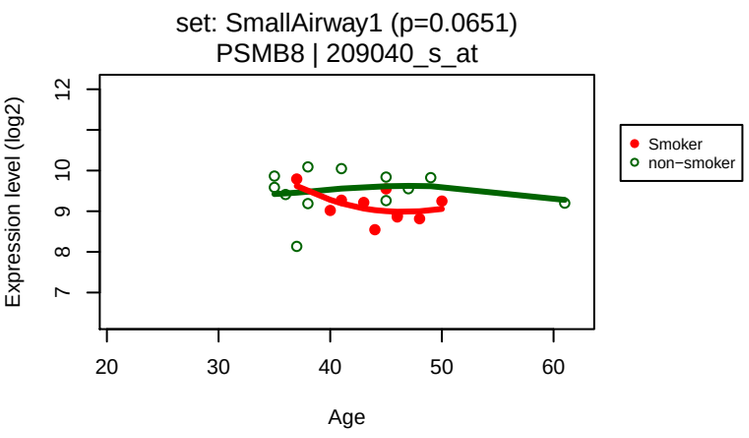
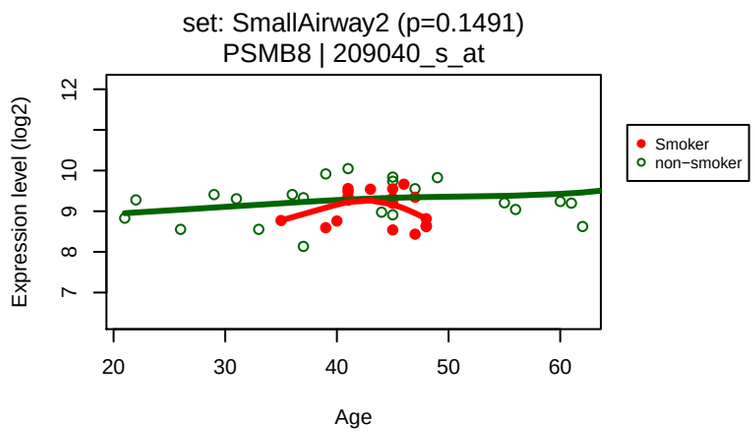
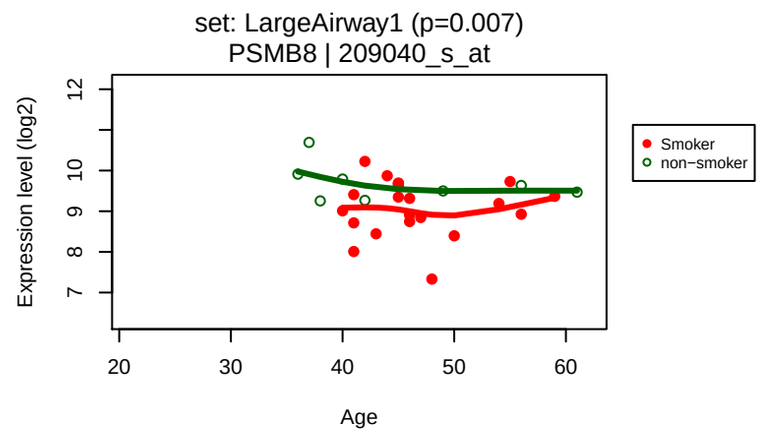

Gene expression of smokers and nonsmokers
gene: PSMB8 | 209040_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0651
p=0.1491
p=0.0278
p=0.007
p=0.5475
p=0.7616
p=0.0136
p=0.3342
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

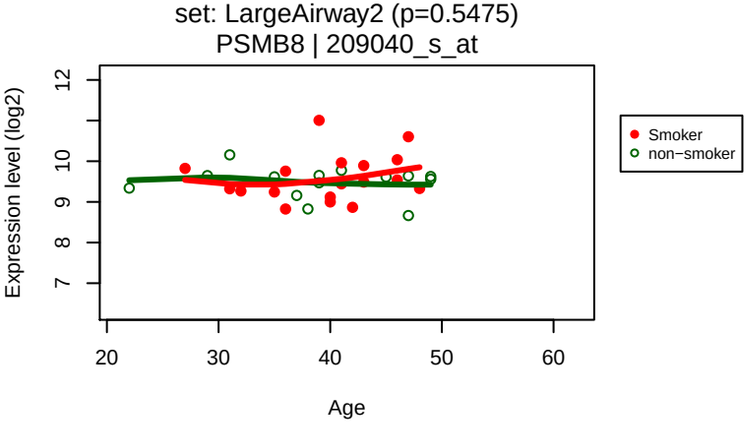

Gene expression ratio (smokers/nonsmokers)
gene: PSMB8 | 209040_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0651
p=0.1491
p=0.0278
p=0.007
p=0.5475
p=0.7616
p=0.0136
p=0.3342
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

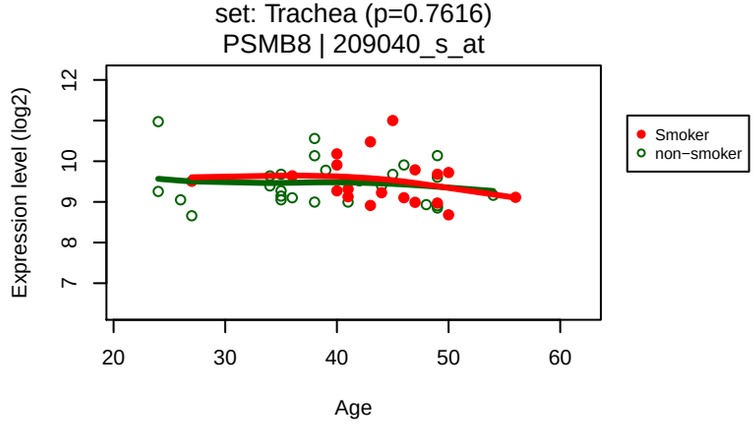

Gene expression ratio in Males
gene: PSMB8 | 209040_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0215
p=0.2895
p=0.739
p=0.1935
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSMB8 | 209040_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5652
p=0.4504
p=0.4547
p=0.8155
SmallAirway2
LargeAirway2
Trachea
Macrophage2

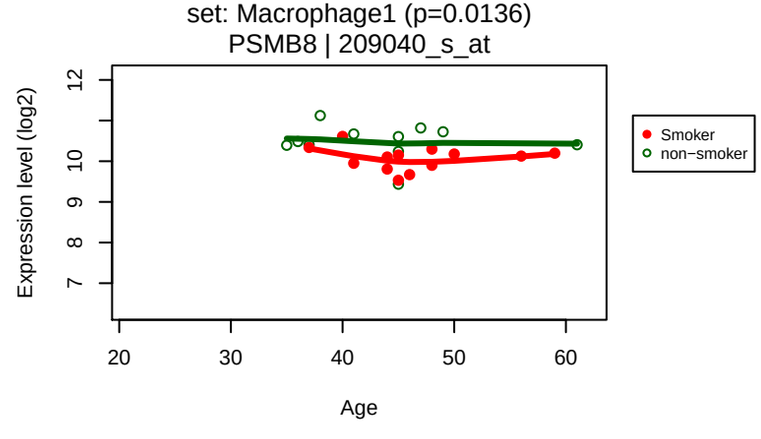

set: SmallAirway1 (p=0.0651)
PSMB8 | 209040_s_at
Expression level (log2)
Age
Smoker
non−smoker

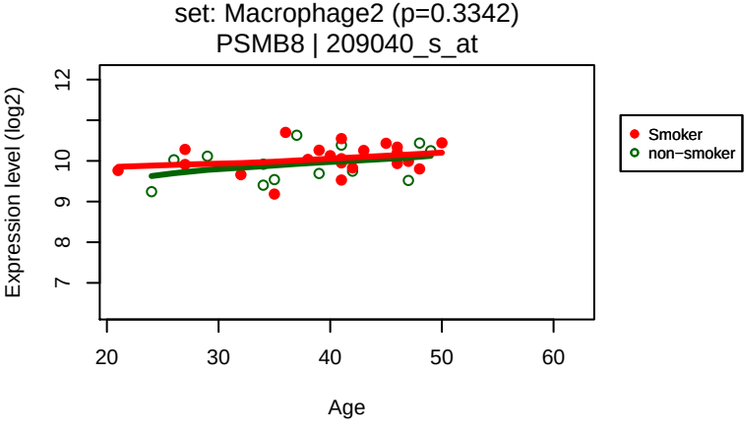

set: SmallAirway2 (p=0.1491)
PSMB8 | 209040_s_at
Expression level (log2)
Age
Smoker
non−smoker

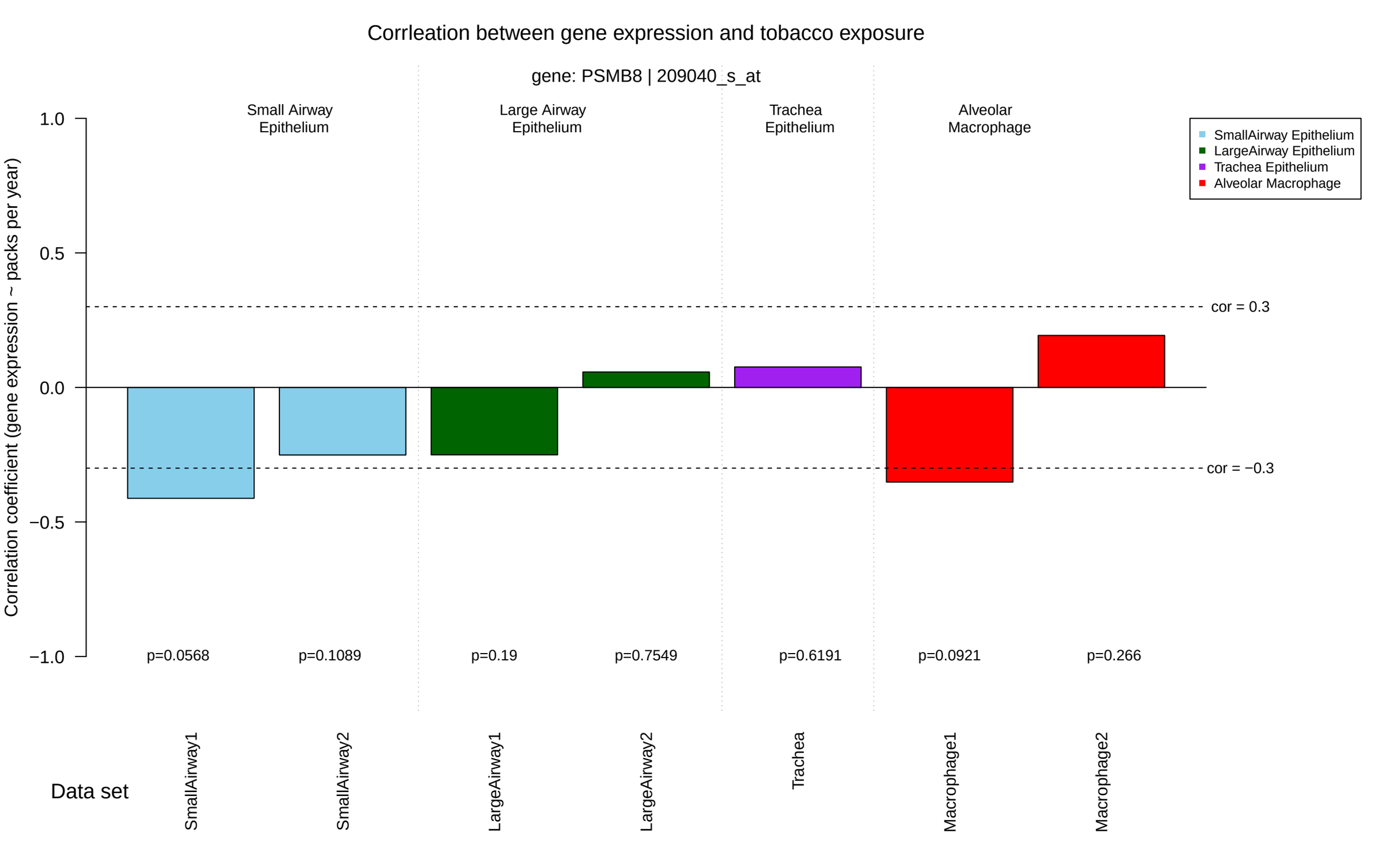

set: LargeAirway1 (p=0.007)
PSMB8 | 209040_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5475)
PSMB8 | 209040_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7616)
PSMB8 | 209040_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0136)
PSMB8 | 209040_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3342)
PSMB8 | 209040_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSMB8 | 209040_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0568
p=0.1089
p=0.19
p=0.7549
p=0.6191
p=0.0921
p=0.266
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2